

Data Overview

The dataset consists of 583 observations and 11 variables related to liver function tests. The key features include demographic information (Age, Gender) and biochemical markers commonly used to assess liver health:

- **Total Bilirubin & Direct Bilirubin** – Indicators of liver's ability to process waste.
- **ALP (Alkaline Phosphatase)** – Enzyme linked to bile duct function.
- **ALT (Alanine Transaminase) & AST (Aspartate Transaminase)** – Enzymes associated with liver cell damage.
- **Total Proteins & Albumin** – Measures of liver's protein synthesis capability.
- **A/G Ratio (Albumin/Globulin Ratio)** – Marker for liver dysfunction and protein balance.
- **Dataset Column** – (unknown)

In terms of data types, all variables except Gender are stored as either integers or floating-point values, ensuring they are numerically operable for statistical analysis. The Gender column is of an object (categorical) data type, which will require encoding for numerical processing in certain analytical methods

Data Cleaning and Transformation

To ensure the dataset is ready for analysis, several data cleaning and preprocessing steps were performed using Python as follows.

1. Data Integrity Issues

- **Column Name Errors:** The variable "Total Protiens" was misspelled and corrected to "Total Proteins."
- **Irrelevant Column:** The column "Dataset" contained no useful information and was removed.

2. Data Standardization

- The "Gender" column was converted to a categorical data type to optimize storage and processing.

- All numerical variables, except for "Age" (which remained as an integer), were standardized to floating-point (float) data types for consistency in calculations.

3. Handling Missing Values

- Missing values were identified in Direct Bilirubin (1), ALP (4), AST (1), Total Proteins (1), and A/G Ratio (5).
- Imputation Strategy: The K-Nearest Neighbors (KNN) method was applied to impute missing values, leveraging patterns in the data to estimate realistic replacements.

4. Handling Duplicate Entries

- 13 duplicate rows were detected and removed to prevent redundancy and ensure data integrity.

5. Handling Outliers

- The dataset had 5 columns which were heavily skewed to the right. This indicated extreme outlier with large data entry figures.
- I explored various methods for handling outliers, but none yielded satisfactory results. Removing the outliers introduced inconsistencies in the expected correlations between columns.
- Therefore I proceeded with analysis including the outliers.

6. Feature Engineering

- Two additional columns, Age Group and AST/ALT ratio, was created to enhance analysis
- The AST/ALT ratio serves as a key diagnostic marker for identifying conditions that cause direct liver cell damage (e.g., alcoholic liver disease, cirrhosis). It was computed as $AST \div ALT$, allowing for a more refined assessment of liver health.
- The age groupings were Infants (1 - 12), Adolescents (13 - 19), young adults (20 - 39), middle-aged adults (40 - 59) and seniors (60+). This segmentation helped identify trends and disparities in liver health across different life stages.